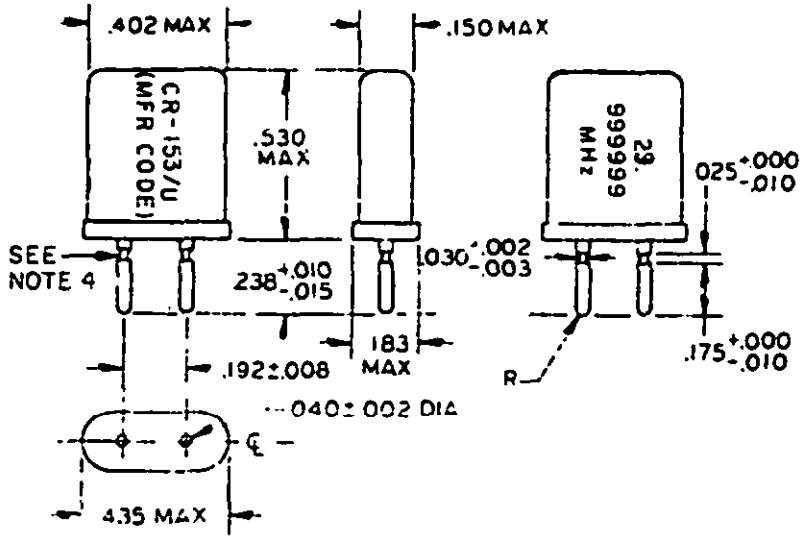
MILITARY SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR-153/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the crystal unit described herein shall consist of this specification and the latest issue of MIL-C-3098.

Pertinent characteristics: 10.0 to 30.0 MHz; fundamental; noncontrolled; antiresonance; nickel electroplated bonding.



INCHES	MM	INCHES	MM
.002	.05	.150	3.81
.003	.08	.175	4.45
.008	.20	.183	4.65
.010	.25	.192	4.88
.015	.38	.238	6.05
.025	.64	.402	10.21
.030	.76	.435	11.05
.040	1.02	.530	13.46

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1 inch = 25.4 mm.
3. The type designation, frequency, and code designation characters shall be not less than .038 (.97 mm) in height and not less than .027 (.69 mm) in width.
4. The pin undercut may be omitted.

FIGURE 1. Crystal unit - CR-153/U

(B) denotes changes

FSC 5955

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TABLE I. Requirements.

Requirement	Specified value or condition
Holder 1/	HC-42-2/U (MIL-H-10056/29)
Dimensions, marking, and configuration	See figure 1
Frequency range	10 to 30 MHz, inclusive
Frequency tolerance:	
Operating temperature	±10 ppm
Equivalent resistance	10 ohms, maximum
Antiresonance, load capacitance	20 ±0.1 pF
Mode of oscillation	Fundamental
Temperature range:	
Operable	0°C to 60°C
Operating	-55°C to +105°C
Test set	TS-683/TSM
Rated drive level	150 microwatts, ±10 percent
C_1	24 pF, ±10 percent
C_0/C_1 ratio	245, ±20 percent
Frequency stability $\frac{\Delta F}{F}$	≤ 3 ppm over the temperature range 0° to 60°C, when measured at series resonance
Shock (specified pulse): 2/	
Frequency change permitted	±0.5 ppm
Equivalent-resistance change permitted	±10 percent or 1 ohm, whichever is greater
Vibration: 3/	
Frequency change permitted	±0.5 ppm
Equivalent-resistance change permitted	±10 percent or 1 ohm, whichever is greater
Temperature run:	
Frequency change permitted	±0.5 ppm
Aging:	
Frequency change permitted	≤ .01 ppm/week at +40°C

1/ Mount shall be subminiature clip mount. Resonator shall be bonded into the subminiature clip mount by the nickel electroplated bonding method. Reference U.S. Army Command Technical Report ECOM 3452, dated August 1971, "Nickel Electrobonded Quartz Blanks".

2/ Method 213 of MIL-STD-202, test condition E, except duration shall be 11 milliseconds. Measurements before and after test - Frequency mesured at +40°C ±1°C; and equivalent resonant resistance measured at 27°C ±1°C.

3/ Method 201 of MIL-STD-202.

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(B) QUALITY ASSURANCE PROVISIONS:

Qualification inspection: Not applicable for this specification.

Quality conformance inspection: Group A tests of MIL-C-3098 shall be applicable.

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - MI, AR, SM
Navy - SH
Air Force - 11, 99
DLA - ES

User activities:
Navy - AS, CG, MC, OS
Air Force - 17, 19

Agent:
DLA - ES

Preparing activity:
Army - ER

(Project 5955-0617-11)

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